

Linear Algebra Recitation

Week 10

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Eigenvalues and eigenvectors

Definition (Eigenvalue, eigenvector)

$$A \in \mathbb{R}^{n \times n}$$

(λ, x) is an eigenpair (eigenvalue, eigenvector) of the matrix A if:

$$x \neq \vec{0}, \quad Ax = \lambda x$$

- An eigenvector x is the "sweet spot" of the matrix A : Ax does not rotate x . It only scales x by λ .



- If x is **an** eigenvector corresponding to λ , what else is an eigenvector?

$$Ax = \lambda x$$

★ infinitely many e.vecs.

$$cAx = c\lambda x$$

How many Lin. indep e.vecs
for A ?

$$A(cx) = \lambda(cx)$$

Exercises

- If x is an eigenvector of A , how would you compute $A^{100}x$?
Multiplying a moderate-sized ($n \sim 10^4$) matrix 100 times will even take a computer a very long time!

$$Ax = \lambda x$$

$$\begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} w_0 \\ d_0 \end{bmatrix} = \begin{bmatrix} w_1 \\ d_1 \end{bmatrix}$$

$$A^2x = \lambda \underbrace{Ax} = \lambda^2 x$$

$$\begin{bmatrix} w_{100} \\ d_{100} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}^{100} \begin{bmatrix} w_0 \\ d_0 \end{bmatrix}$$

$$A^{100}x = \lambda^{100}x \quad \text{if we have 2 lin. indep e. vecs}$$

$$\begin{bmatrix} w_0 \\ d_0 \end{bmatrix} = C_1 x_1 + C_2 x_2$$

$$\begin{bmatrix} w_{100} \\ d_{100} \end{bmatrix} = C_1 \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}^{100} x_1 + C_2 \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}^{100} x_2 = C_1 \lambda_1^{100} x_1 + C_2 \lambda_2^{100} x_2$$

How to find eigenvalue λ ?

$$Bx = 0$$

- (Definition) $Ax = \lambda x \implies (A - \lambda I)x = 0$.

- When does a nonzero x exist?

$N(A - \lambda I)$ is non-trivial $\Leftrightarrow$ not injective
 $\Leftrightarrow \det(A - \lambda I) = 0$

- What is $\det(A - \lambda I)$ called?

Characteristic polynomial

$$\begin{bmatrix} a-\lambda & b & c \\ d & e-\lambda & f \\ g & h & i-\lambda \end{bmatrix}$$

- This is often denoted as $p_A(\lambda) = \det(A - \lambda I)$.
- For $A \in \mathbb{R}^{n \times n}$, what is the degree of $p_A(\lambda)$?

n

- How can we find the eigenvalues from $p_A(\lambda)$?

e. vals are the roots of $p_A(\lambda)$

How to find eigenvector x corresponding to λ ?

- Once we know an eigenvalue λ of A , how can we find the associated eigenvectors?
- Recall, an eigenpair must satisfy $(A - \lambda I)x = 0$. What can be said about the null space of $A - \lambda I$?

$$Bx = 0$$

$$(A - \lambda I)x = 0$$

find basis for $N(A - \lambda I)$

if $N(A - \lambda I) = \text{Span}\{\underline{x_1}, x_2\}$

Thm) $Ax_3 = \lambda_1 x_3$, $Ax_4 = \lambda_2 x_4$

$\lambda_1 \neq \lambda_2 \Rightarrow x_1, x_2$ Lin. indep.

Def) if λ has multiplicity 1 in $P_A(\lambda)$: non-degenerate e. val.

Exercises

- Let $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Find the eigenvalues of A , and associated eigenvectors.

① form $P_A(\lambda) \Rightarrow$ find e. vals

② find e. vecs.

$$P_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{bmatrix} = (\lambda-1)^2 - 1$$

$$\text{i) } \lambda_1 = 0 \Rightarrow (A - \lambda_1 I)x = 0 \quad \text{ii) } \lambda_2 = 2 \Rightarrow (A - \lambda_2 I)x = 0$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} x = 0$$

$$x_1 - x_2 = 0$$

$$N(A - \lambda_1 I) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

$$x^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} x = 0$$

$$x_1 + x_2 = 0$$

$$N(A - \lambda_2 I) = \text{Span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

$$x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Exercises Diagonal / U. triangular / L. triangular
 $\Rightarrow$ diagonal entries are e. vals, and det is the product of diag.

- Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Find the eigenvalues of A , and associated eigenvectors.

$$P_A(\lambda) = \det(A - \lambda I) = (1 - \lambda)^2$$

$$(A - I)x = 0$$

$$\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} x = 0$$

$$x_2 = 0 \Rightarrow N(A - I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

- It is generally easy to tell if a given vector is an eigenvector or not:
- Without prior knowledge, are the vectors $x_1 = (1, 1)^T$ and $x_2 = (-2, 0)^T$ eigenvectors of A above? If so, what are the associated eigenvalues?

$$Ax_1 = A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Ax_2 = A \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

A special case:

- A has a zero eigenvalue $\iff Ax = 0$

$$\iff N(A) \text{ non trivial}$$

$$\iff \det(A) = 0$$

Similar matrices $AX = XB \Rightarrow [Ax_1 | \dots | Ax_n] = [b_1x_1 | \dots | b_nx_n]$
 $[x_1 | \dots | x_n] \begin{bmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{bmatrix}$

Definition

A and B are similar matrices if: there exists an invertible X

$$A = XBX^{-1}$$

- Similarity preserves characteristic polynomial
- Therefore, similar matrices must have same eigenvalues.

Exercises

- Are $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$ similar? Answer in 3 seconds.
- In general, it is not trivial to tell whether two matrices are similar or not, especially when two matrices have the same eigenvalues.
- Similar matrices must have same eigenvalues, but matrices with same eigenvalues need not be similar!
- e.g. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ are not similar because they do not have the same Jordan forms (a topic that may be covered later).

Diagonalization and eigen (spectral) decomposition

A is diagonalizable if there is an invertible matrix C and a diagonal matrix B such that

$$A = CBC^{-1} \leftarrow A \text{ must be square matrix}$$

$$AC = CB$$

$$A[C_1 \dots C_n] = CB = C \begin{bmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{bmatrix}$$

$$[AC_1 \dots AC_n] = [b_1 C_1 \dots b_n C_n]$$

$$AC_k = b_k C_k \Rightarrow (b_k, C_k) \text{ are e. pairs.}$$

Not diagonalizable when A does not have n lin. indep e. vecs.

$$A = CBC^{-1} : \begin{matrix} \text{spectral} \\ \text{eigen} \end{matrix} \text{ decomp. } B = C^{-1}AC : \text{diagonalization}$$

Diagonalization and eigen (spectral) decomposition

- When is a matrix Diagonalizable? When A has n lin. indep evecs.
(to make C invertible)

Some Sufficient Conditions for A to be diagonal-

① A has n distinct e.vals.

Lemma: e.vectors corresponding to distinct e.vals are lin. indep.

② A is symmetric.

Diagonalization and eigen (spectral) decomposition

a) Compute the eigenvalues of the matrix

$$\textcircled{1} \text{ Given } P_A(\lambda) = \det(A - \lambda I)$$

$$= \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$
$$= \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$
$$\lambda_1 = 3, \lambda_2 = 1$$

b) Find a matrix C such that

$$B = C^{-1}AC$$

where B has only diagonal entries.

$$C = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\textcircled{1} \lambda_1 = 3 \Rightarrow (A - 3I)x = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0$$

$$\Rightarrow x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\textcircled{2} \lambda_2 = 1 \Rightarrow (A - I)x = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0$$

$$\Rightarrow x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$